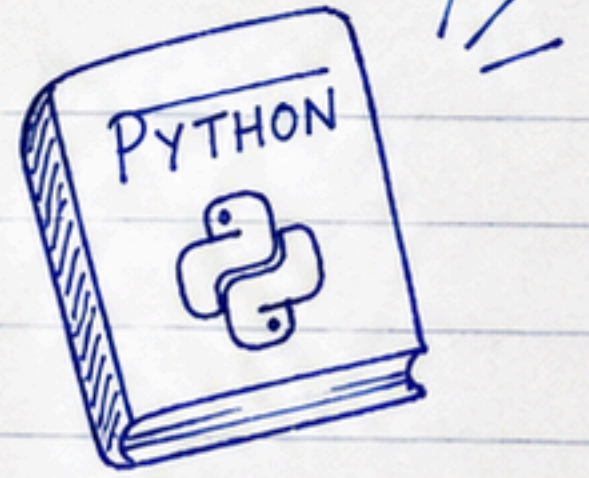




# CHAPTER 03

## Strings



- String is a data type in python.
- String is a sequence of characters enclosed in quotes.
- We can primarily write a string in these three ways.

```
a = 'harry'      # Single quoted string
b = "harry"      # Double quoted string
c = """harry"""  # Triple quoted string
```

Single quoted string

'harry'



Double quoted string

"harry"



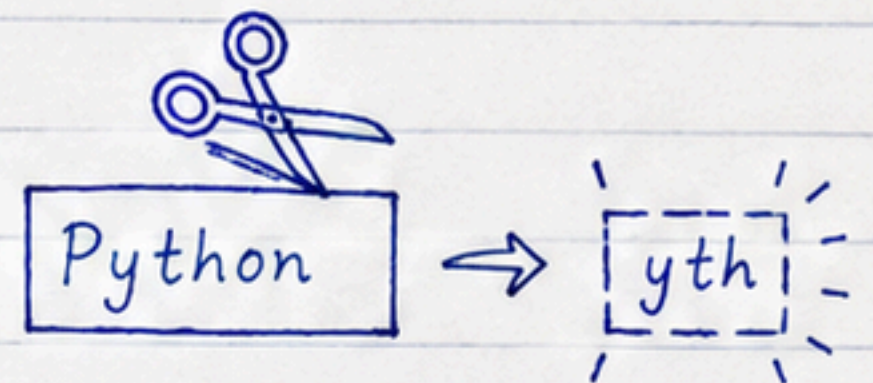
Triple quoted string

"""harry"""



## String Slicing

- A string in python can be sliced for getting a part of the strings.



Consider the following string:

s = "Python"

|         |    |    |    |    |    |    |
|---------|----|----|----|----|----|----|
| Index → | 0  | 1  | 2  | 3  | 4  | 5  |
|         | P  | y  | t  | h  | o  | n  |
| Index → | -6 | -5 | -4 | -3 | -2 | -1 |

The index in a string starts from 0 to (length - 1) in Python.



In order to slice a string, we use the following Syntax:

string\_name[start : end : step]

Starting Index

Ending Index

Step Size





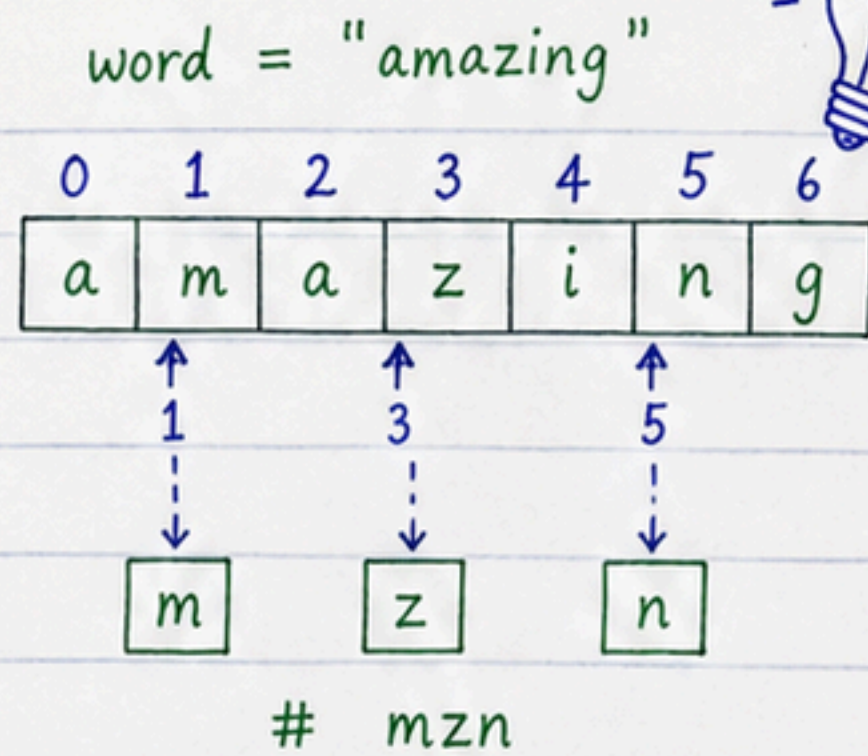


## Slicing with Skip Value



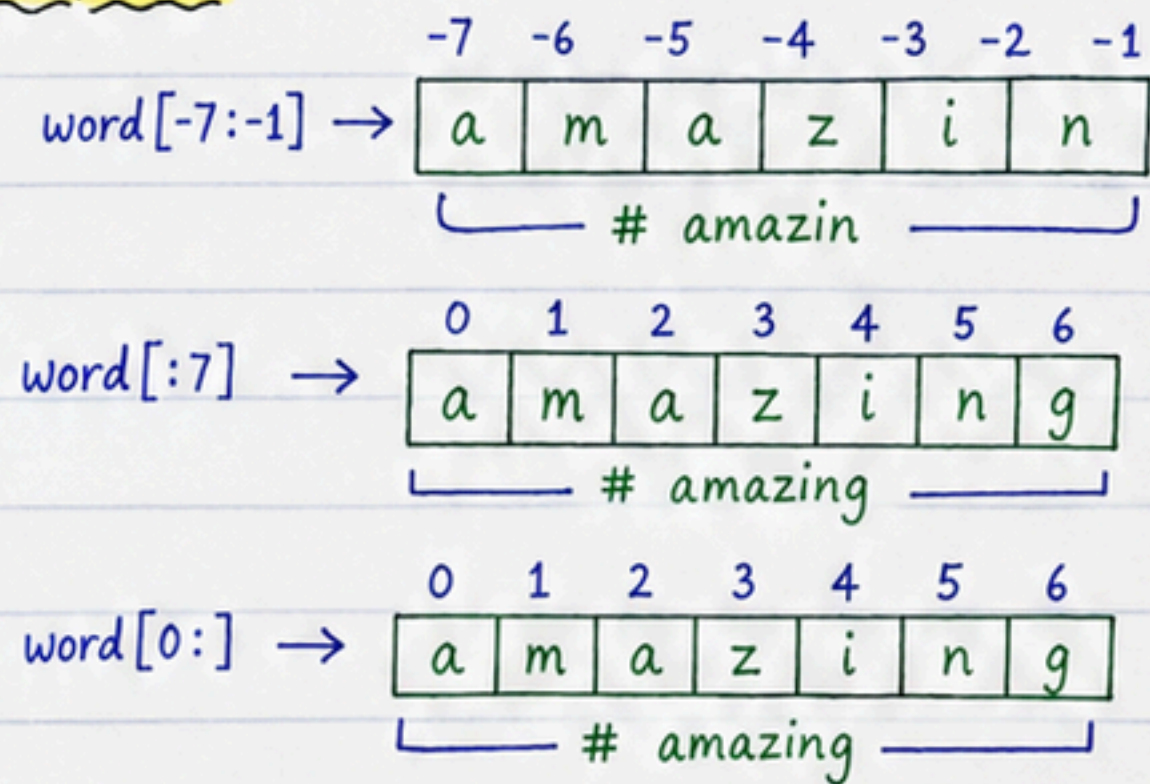
- We can provide a skip value as a part of our slice like this:

```
word = "amazing"
word[1:6:2] # mzn
```



## Other advanced slicing techniques:

```
word = "amazing"
word[-7:-1] # amazin
word[:7] # amazing
word[0:] # amazing
```

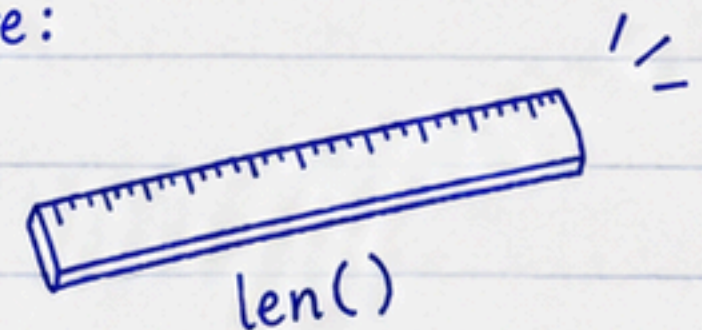


## String Functions

Some commonly used functions to manipulate strings are:

- len()** returns the length of the string.

```
str = "micro"
print(len(str)) # Output: 5
```



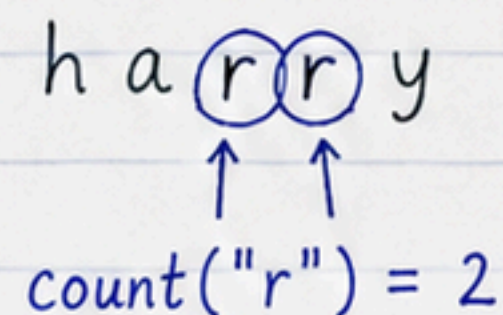
- endswith()** checks if a string ends with given text.

```
str = "Micro"
print(str.endswith("rry")) # Output: True
```



- count()** counts total occurrences of a character.

```
str = "harry"
count = str.count("r")
print(count) # Output: 2
```



- capitalize()** capitalizes the first character.

```
str = "Micro"
capitalized = str.capitalize()
print(capitalized) # Output: Micro
```

"Micro"

↓

"Micro"







## String Functions Continued

5. **find()** returns the index of first occurrence.

```
str = "harry"
index = str.find("rr")
print(index)    # Output: 2
```



Finds the first occurrence and returns its index.



6. **replace(old word, new word)** replaces the old word with the new word in the string.

```
str = "harry"
replaced = str.replace("r", "l")
print(replaced)    # Output: hally
```

"harry"  $\xrightarrow[r \rightarrow l]{\text{replace}}$  "hally"



## Escape Sequence Characters

Sequence of characters after backslash "\" are called **Escape Sequence** characters.

These characters represent one special character inside strings.

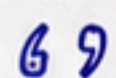
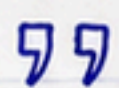


**Example:**

| Escape Sequence | Represents   | Example                                            |
|-----------------|--------------|----------------------------------------------------|
| \n              | New Line     | print("Hello\nWorld")<br>Output: Hello<br>World    |
| \t              | Tab          | print("A\tB")<br>Output: A    B                    |
| \\              | Backslash    | print("C:\\Python")<br>Output: C:\Python           |
| \"              | Double Quote | print("He said, \"Hi!\")<br>Output: He said, "Hi!" |
| \'              | Single Quote | print('It\'s fun')<br>Output: It's fun             |



A ----> B



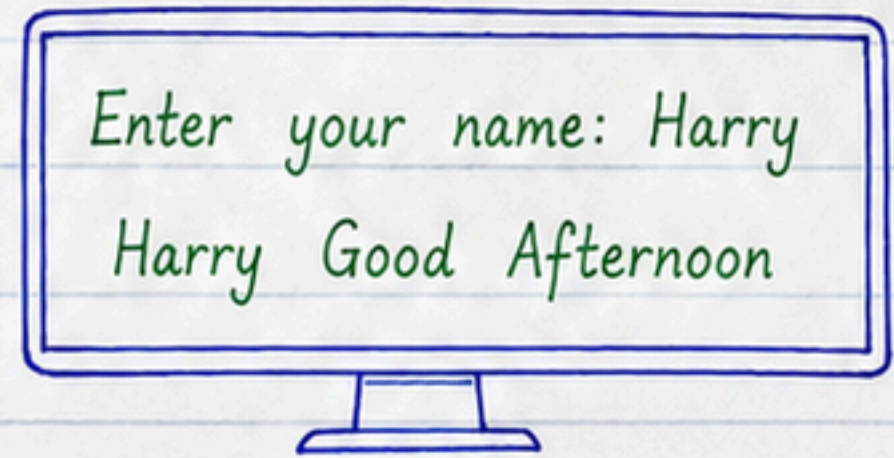




# Practice Set

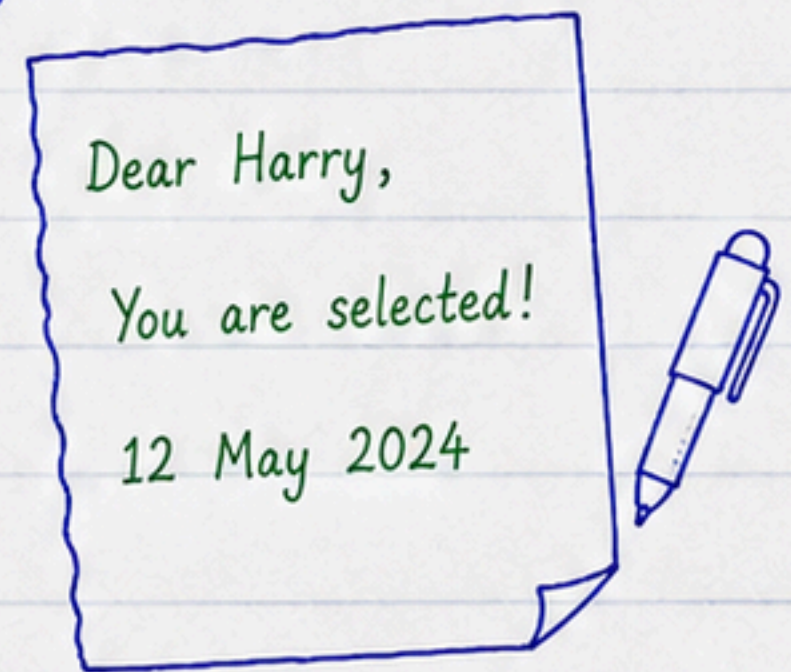


1. Write a python program to display a user entered name followed by Good Afternoon using input() function.



2. Write a program to fill in a letter template given below with name and date.

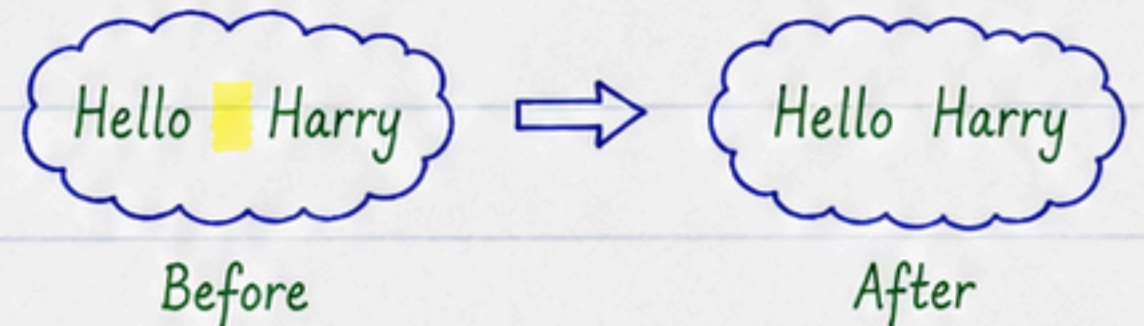
```
letter = '''
Dear <|Name|>,
You are selected!
<|Date|>
'''
```



3. Write a program to detect double space in a string.



4. Replace the double space from problem 3 with single spaces.



5. Write a program to format the following letter using escape sequence characters.

```
letter = "Dear Micro, this python
course is nice. Thanks!"
```

